

Geospatial Analysis using R

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Setting up R environment with essential packages.

- Installing Required Packages

```
# install.packages(c("sf", "terra", "tmap", "leaflet", "ggplot2", "tidyverse", "spData", "ray  
shader", "viridis"))
```

- Loading the packages

```
library(sf)  
library(terra)  
library(tmap)  
library(leaflet)  
library(ggplot2)  
library(rayshader)  
library(viridis)  
library(tidyverse)  
library(spData)
```

```
head(world)
```

```
## Simple feature collection with 6 features and 10 fields  
## Geometry type: MULTIPOLYGON  
## Dimension:      XY  
## Bounding box:   xmin: -180 ymin: -18.28799 xmax: 180 ymax: 83.23324  
## Geodetic CRS:   WGS 84  
## # A tibble: 6 × 11  
##   iso_a2 name_long continent region_un subregion type area_km2 pop lifeExp  
##   <chr>  <chr>      <chr>   <chr>   <chr>   <chr>   <dbl>   <dbl>   <dbl>  
## 1 FJ    Fiji       Oceania Oceania Melanesia Sove... 1.93e4 8.86e5 70.0  
## 2 TZ    Tanzania Africa Africa Eastern ... Sove... 9.33e5 5.22e7 64.2  
## 3 EH    Western S... Africa Africa Northern... Inde... 9.63e4 NA      NA  
## 4 CA    Canada      North Am... Americas Northern... Sove... 1.00e7 3.55e7 82.0  
## 5 US    United St... North Am... Americas Northern... Coun... 9.51e6 3.19e8 78.8  
## 6 KZ    Kazakhstan Asia      Asia      Central ... Sove... 2.73e6 1.73e7 71.6  
## # i 2 more variables: gdpPercap <dbl>, geom <MULTIPOLYGON [°]>
```

```
tail(world)
```

```
## Simple feature collection with 6 features and 10 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: -61.95 ymin: 3.509172 xmax: 35.29801 ymax: 46.17173
## Geodetic CRS: WGS 84
## # A tibble: 6 × 11
##   iso_a2 name_long continent region_un subregion type area_km2 pop lifeExp
##   <chr> <chr> <chr> <chr> <chr> <chr> <dbl> <dbl> <dbl>
## 1 MK Macedonia Europe Europe Southern... Sove... 25062. 2.08e6 75.4
## 2 RS Serbia Europe Europe Southern... Sove... 76389. 7.13e6 75.3
## 3 ME Montenegro Europe Europe Southern... Sove... 13444. 6.22e5 76.7
## 4 XK Kosovo Europe Europe Southern... Sove... 11230. 1.82e6 71.1
## 5 TT Trinidad a... North Am... Americas Caribbean Sove... 7738. 1.35e6 70.4
## 6 SS South Sudan Africa Africa Eastern ... Sove... 624909. 1.15e7 55.8
## # i 2 more variables: gdpPercap <dbl>, geom <MULTIPOLYGON [°]>
```

```
sample(world)
```

```
## Simple feature collection with 177 features and 10 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: -180 ymin: -89.9 xmax: 180 ymax: 83.64513
## Geodetic CRS: WGS 84
## # A tibble: 177 × 11
##   area_km2 geom iso_a2 name_long continent lifeExp
##   <dbl> <MULTIPOLYGON [°]> <chr> <chr> <chr> <dbl>
## 1 19290. (((-180 -16.55522, -179.9174 -1... FJ Fiji Oceania 70.0
## 2 932746. (((33.90371 -0.95, 31.86617 -1... TZ Tanzania Africa 64.2
## 3 96271. (((-8.66559 27.65643, -8.817828... EH Western ... Africa NA
## 4 10036043. (((-132.71 54.04001, -133.18 54... CA Canada North Am... 82.0
## 5 9510744. (((-171.7317 63.78252, -171.791... US United S... North Am... 78.8
## 6 2729811. (((87.35997 49.21498, 86.82936 ... KZ Kazakhst... Asia 71.6
## 7 461410. (((55.96819 41.30864, 57.09639 ... UZ Uzbekist... Asia 71.0
## 8 464520. (((141.0002 -2.600151, 141.0171... PG Papua Ne... Oceania 65.2
## 9 1819251. (((104.37 -1.084843, 104.0108 -... ID Indonesia Asia 68.9
## 10 2784469. (((-68.63401 -52.63637, -68.633... AR Argentina South Am... 76.3
## # i 167 more rows
## # i 5 more variables: gdpPercap <dbl>, pop <dbl>, subregion <chr>, type <chr>,
## # region_un <chr>
```

```
glimpse(world)
```

```
## Rows: 177
## Columns: 11
## $ iso_a2      <chr> "FJ", "TZ", "EH", "CA", "US", "KZ", "UZ", "PG", "ID", "AR", ...
## $ name_long   <chr> "Fiji", "Tanzania", "Western Sahara", "Canada", "United Stat...
## $ continent   <chr> "Oceania", "Africa", "Africa", "North America", "North Ameri...
## $ region_un   <chr> "Oceania", "Africa", "Africa", "Americas", "Americas", "Asia...
## $ subregion   <chr> "Melanesia", "Eastern Africa", "Northern Africa", "Northern ...
## $ type        <chr> "Sovereign country", "Sovereign country", "Indeterminate", "...
## $ area_km2    <dbl> 19289.97, 932745.79, 96270.60, 10036042.98, 9510743.74, 2729...
## $ pop         <dbl> 885806, 52234869, NA, 35535348, 318622525, 17288285, 3075770...
## $ lifeExp     <dbl> 69.96000, 64.16300, NA, 81.95305, 78.84146, 71.62000, 71.039...
## $ gdpPercap   <dbl> 8222.2538, 2402.0994, NA, 43079.1425, 51921.9846, 23587.3375...
## $ geom        <MULTIPOLYGON [°]> MULTIPOLYGON (((-180 -16.55..., MULTIPOLYGON ((...
```

```
class(world)
```

```
## [1] "sf"           "tbl_df"      "tbl"         "data.frame"
```

```
names(world)
```

```
## [1] "iso_a2"      "name_long"   "continent"   "region_un"   "subregion"   "type"
## [7] "area_km2"    "pop"         "lifeExp"     "gdpPercap"   "geom"
```

```
str(world)
```

```
## sf [177 × 11] (S3: sf/tbl_df/tbl/data.frame)
## $ iso_a2 : chr [1:177] "FJ" "TZ" "EH" "CA" ...
## $ name_long: chr [1:177] "Fiji" "Tanzania" "Western Sahara" "Canada" ...
## $ continent: chr [1:177] "Oceania" "Africa" "Africa" "North America" ...
## $ region_un: chr [1:177] "Oceania" "Africa" "Africa" "Americas" ...
## $ subregion: chr [1:177] "Melanesia" "Eastern Africa" "Northern Africa" "Northern America"
...
## $ type : chr [1:177] "Sovereign country" "Sovereign country" "Indeterminate" "Sovereign
country" ...
## $ area_km2 : num [1:177] 19290 932746 96271 10036043 9510744 ...
## $ pop : num [1:177] 8.86e+05 5.22e+07 NA 3.55e+07 3.19e+08 ...
## $ lifeExp : num [1:177] 70 64.2 NA 82 78.8 ...
## $ gdpPercap: num [1:177] 8222 2402 NA 43079 51922 ...
## $ geom :sfc_MULTIPOLYGON of length 177; first list element: List of 3
## ..$ :List of 1
## .. ..$ : num [1:5, 1:2] -180 -180 -180 -180 -180 ...
## ..$ :List of 1
## .. ..$ : num [1:9, 1:2] 178 178 177 177 178 ...
## ..$ :List of 1
## .. ..$ : num [1:8, 1:2] 180 180 179 179 179 ...
## ..- attr(*, "class")= chr [1:3] "XY" "MULTIPOLYGON" "sfg"
## - attr(*, "sf_column")= chr "geom"
## - attr(*, "agr")= Factor w/ 3 levels "constant","aggregate",..: NA NA NA NA NA NA NA NA NA
A
## ..- attr(*, "names")= chr [1:10] "iso_a2" "name_long" "continent" "region_un" ...
```

```
summary(world)
```

```
##      iso_a2          name_long      continent      region_un
## Length:177      Length:177      Length:177      Length:177
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
##
##      subregion          type          area_km2          pop
## Length:177      Length:177      Min.   :   2417      Min.   :5.630e+04
## Class :character Class :character 1st Qu.:  46185      1st Qu.:3.755e+06
## Mode  :character Mode  :character Median : 185004      Median :1.040e+07
##                                     Mean  :  832558      Mean   :4.282e+07
##                                     3rd Qu.: 621860      3rd Qu.:3.075e+07
##                                     Max.   :17018507      Max.   :1.364e+09
##                                     NA's    :10
##
##      lifeExp      gdpPercap          geom
## Min.   :50.62      Min.   :   597.1      MULTIPOLYGON :177
## 1st Qu.:64.96      1st Qu.:  3752.4      epsg:4326     :  0
## Median :72.87      Median : 10734.1      +proj=long... :  0
## Mean   :70.85      Mean   : 17106.0
## 3rd Qu.:76.78      3rd Qu.: 24232.7
## Max.   :83.59      Max.   :120860.1
## NA's   :10        NA's   :17
```

```
view(world)
```

```
sf::st_crs(world)
```

```
## Coordinate Reference System:
##   User input: EPSG:4326
##   wkt:
##   GEOGCRS["WGS 84",
##     DATUM["World Geodetic System 1984",
##       ELLIPSOID["WGS 84",6378137,298.257223563,
##         LENGTHUNIT["metre",1]],
##       PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##     AXIS["geodetic latitude (Lat)",north,
##       ORDER[1],
##       ANGLEUNIT["degree",0.0174532925199433]],
##     AXIS["geodetic longitude (Lon)",east,
##       ORDER[2],
##       ANGLEUNIT["degree",0.0174532925199433]],
##     USAGE[
##       SCOPE["Horizontal component of 3D system."],
##       AREA["World."],
##       BBOX[-90,-180,90,180]],
##     ID["EPSG",4326]]
```

```
# View geometry type
st_geometry_type(world) |> table()
```

```
##
##          GEOMETRY          POINT          LINESTRING          POLYGON
##          0                0                0                0
##      MULTIPOINT  MULTILINESTRING  MULTIPOLYGON  GEOMETRYCOLLECTION
##          0                0                177                0
##  CIRCULARSTRING  COMPOUNDCURVE    CURVEPOLYGON    MULTICURVE
##          0                0                0                0
##      MULTISURFACE      CURVE          SURFACE  POLYHEDRALSURFACE
##          0                0                0                0
##          TIN           TRIANGLE
##          0                0
```

```
# To view number of nulls per column
colSums(is.na(world))
```

```
##   iso_a2 name_long continent region_un subregion   type area_km2   pop
##     2      0         0         0         0       0      0      10
## lifeExp gdpPercap      geom
##    10      17         0
```

```
world %>% filter(is.na(gdpPercap))
```

```
## Simple feature collection with 17 features and 10 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -180 ymin: -89.9 xmax: 180 ymax: 83.64513
## Geodetic CRS:   WGS 84
## # A tibble: 17 × 11
##   iso_a2 name_long continent region_un subregion type area_km2      pop lifeExp
## * <chr> <chr>      <chr>    <chr>    <chr>    <chr>    <dbl>    <dbl>    <dbl>
## 1 EH      Western ... Africa    Africa    Northern... Inde...    9.63e4 NA      NA
## 2 SO      Somalia    Africa    Africa    Eastern ... Sove...    4.84e5 1.35e7 55.5
## 3 FK      Falkland... South Am... Americas    South Am... Depe...    1.64e4 NA      NA
## 4 NO      Norway     Europe    Europe    Northern... Sove...    3.98e5 NA      NA
## 5 GL      Greenland North Am... Americas    Northern... Coun...    2.21e6 5.63e4 NA
## 6 TF      French S... Seven se... Seven se... Seven se... Depe...    1.16e4 NA      NA
## 7 FR      France     Europe    Europe    Western ... Coun...    6.45e5 NA      NA
## 8 CU      Cuba       North Am... Americas    Caribbean Sove...    1.15e5 1.14e7 79.4
## 9 KP      Dem. Rep... Asia      Asia      Eastern ... Sove...    1.26e5 2.51e7 71.2
## 10 SY     Syria      Asia      Asia      Western ... Sove...    1.85e5 1.92e7 69.8
## 11 NC     New Cale... Oceania    Oceania    Melanesia Depe...    2.32e4 2.68e5 77.3
## 12 TW     Taiwan     Asia      Asia      Eastern ... Sove...    3.43e4 NA      NA
## 13 ER     Eritrea    Africa    Africa    Eastern ... Sove...    1.19e5 NA      64.2
## 14 AQ     Antarcti... Antarcti... Antarcti... Antarcti... Inde...    1.23e7 NA      NA
## 15 <NA>    Northern... Asia      Asia      Western ... Sove...    3.79e3 NA      NA
## 16 DJ     Djibouti   Africa    Africa    Eastern ... Sove...    2.19e4 9.12e5 62.0
## 17 <NA>    Somalila... Africa    Africa    Eastern ... Inde...    1.67e5 NA      NA
## # i 2 more variables: gdpPercap <dbl>, geom <MULTIPOLYGON [°]>
```

```
# See rows where any column has NA
# world[!complete.cases(world), ]
```

```
# Fill NA in gdpPercap with mean
mean(world$gdpPercap, na.rm = TRUE)
```

```
## [1] 17105.99
```

```
# world$gdpPercap[is.na(world$gdpPercap)] <- mean(world$gdpPercap, na.rm = TRUE)
```

```
# Exclude geom column using select()
world %>% select(-geom)
```

```
## Simple feature collection with 177 features and 10 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -180 ymin: -89.9 xmax: 180 ymax: 83.64513
## Geodetic CRS:   WGS 84
## # A tibble: 177 × 11
##   iso_a2 name_long continent region_un subregion type area_km2      pop lifeExp
##   <chr>  <chr>      <chr>    <chr>    <chr>    <chr>    <dbl>    <dbl>    <dbl>
## 1 FJ     Fiji       Oceania  Oceania  Melanesia Sove...  1.93e4  8.86e5    70.0
## 2 TZ     Tanzania  Africa   Africa   Eastern ... Sove...  9.33e5  5.22e7    64.2
## 3 EH     Western ... Africa   Africa   Northern... Inde...  9.63e4  NA        NA
## 4 CA     Canada    North Am... Americas Northern... Sove...  1.00e7  3.55e7    82.0
## 5 US     United S... North Am... Americas Northern... Coun...  9.51e6  3.19e8    78.8
## 6 KZ     Kazakhst... Asia     Asia     Central ... Sove...  2.73e6  1.73e7    71.6
## 7 UZ     Uzbekist... Asia     Asia     Central ... Sove...  4.61e5  3.08e7    71.0
## 8 PG     Papua Ne... Oceania  Oceania  Melanesia Sove...  4.65e5  7.76e6    65.2
## 9 ID     Indonesia Asia     Asia     South-Ea... Sove...  1.82e6  2.55e8    68.9
## 10 AR    Argentina South Am... Americas South Am... Sove...  2.78e6  4.30e7    76.3
## # i 167 more rows
## # i 2 more variables: gdpPercap <dbl>, geom <MULTIPOLYGON [°]>
```

```
world$gdpPercap[world$name_long == "Western Sahara"] <- 2800
```

To filter

```
filter(world, name_long == "Nigeria")
```

```
## Simple feature collection with 1 feature and 10 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: 2.691702 ymin: 4.240594 xmax: 14.57718 ymax: 13.86592
## Geodetic CRS:   WGS 84
## # A tibble: 1 × 11
##   iso_a2 name_long continent region_un subregion type area_km2      pop lifeExp
## * <chr> <chr>      <chr>    <chr>    <chr>    <chr>    <dbl>    <dbl>    <dbl>
## 1 NG     Nigeria  Africa   Africa   Western Af... Sove...  905072. 1.76e8    52.5
## # i 2 more variables: gdpPercap <dbl>, geom <MULTIPOLYGON [°]>
```

```
# filtering using piping
nigeria <- world %>% filter(name_long == "Nigeria")
nigeria
```



```
## Simple feature collection with 1 feature and 10 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: 2.691702 ymin: 4.240594 xmax: 14.57718 ymax: 13.86592
## Geodetic CRS:   WGS 84
## # A tibble: 1 × 11
##   iso_a2 name_long continent region_un subregion  type area_km2    pop lifeExp
## * <chr> <chr>      <chr>      <chr>      <chr>      <chr>    <dbl> <dbl> <dbl>
## 1 NG      Nigeria    Africa    Africa    Western Af... Sove...  905072.  1.76e8   52.5
## # i 2 more variables: gdpPercap <dbl>, geom <MULTIPOLYGON [°]>
```

```
Africa <- world %>% filter(continent == "Africa")
Africa
```

```
## Simple feature collection with 51 features and 10 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -17.62504 ymin: -34.81917 xmax: 51.13387 ymax: 37.34999
## Geodetic CRS:   WGS 84
## # A tibble: 51 × 11
##   iso_a2 name_long continent region_un subregion type area_km2    pop lifeExp
## * <chr> <chr>      <chr>      <chr>      <chr>      <chr>    <dbl> <dbl> <dbl>
## 1 TZ      Tanzania    Africa    Africa    Eastern ... Sove...  932746.  5.22e7   64.2
## 2 EH      Western ... Africa    Africa    Northern... Inde...   96271.  NA      NA
## 3 CD      Democrat... Africa    Africa    Middle A... Sove... 2323492.  7.37e7   58.8
## 4 SO      Somalia     Africa    Africa    Eastern ... Sove...  484333.  1.35e7   55.5
## 5 KE      Kenya     Africa    Africa    Eastern ... Sove...  590837.  4.60e7   66.2
## 6 SD      Sudan       Africa    Africa    Northern... Sove... 1850886.  3.77e7   64.0
## 7 TD      Chad        Africa    Africa    Middle A... Sove... 1271695.  1.36e7   52.2
## 8 ZA      South Af... Africa    Africa    Southern... Sove... 1216401.  5.45e7   61.0
## 9 LS      Lesotho     Africa    Africa    Southern... Sove...   27506.  2.15e6   53.3
## 10 ZW     Zimbabwe   Africa    Africa    Eastern ... Sove...  376328.  1.54e7   59.4
## # i 41 more rows
## # i 2 more variables: gdpPercap <dbl>, geom <MULTIPOLYGON [°]>
```

```
world$continent
```

##	[1]	"Oceania"	"Africa"
##	[3]	"Africa"	"North America"
##	[5]	"North America"	"Asia"
##	[7]	"Asia"	"Oceania"
##	[9]	"Asia"	"South America"
##	[11]	"South America"	"Africa"
##	[13]	"Africa"	"Africa"
##	[15]	"Africa"	"Africa"
##	[17]	"North America"	"North America"
##	[19]	"Europe"	"North America"
##	[21]	"South America"	"Europe"
##	[23]	"North America"	"Seven seas (open ocean)"
##	[25]	"Asia"	"Africa"
##	[27]	"Africa"	"North America"
##	[29]	"South America"	"South America"
##	[31]	"South America"	"South America"
##	[33]	"South America"	"North America"
##	[35]	"North America"	"North America"
##	[37]	"North America"	"North America"
##	[39]	"North America"	"North America"
##	[41]	"South America"	"South America"
##	[43]	"South America"	"Europe"
##	[45]	"South America"	"North America"
##	[47]	"North America"	"North America"
##	[49]	"Africa"	"Africa"
##	[51]	"Africa"	"Africa"
##	[53]	"Africa"	"Africa"
##	[55]	"Africa"	"Africa"
##	[57]	"Africa"	"Africa"
##	[59]	"Africa"	"Africa"
##	[61]	"Africa"	"Africa"
##	[63]	"Africa"	"Africa"
##	[65]	"Africa"	"Africa"
##	[67]	"Africa"	"Africa"
##	[69]	"Africa"	"Africa"
##	[71]	"Africa"	"Africa"
##	[73]	"Africa"	"Africa"
##	[75]	"Africa"	"Africa"
##	[77]	"Asia"	"Asia"
##	[79]	"Africa"	"Asia"
##	[81]	"Africa"	"Africa"
##	[83]	"Africa"	"Asia"
##	[85]	"Asia"	"Asia"
##	[87]	"Asia"	"Asia"
##	[89]	"Asia"	"Oceania"
##	[91]	"Asia"	"Asia"
##	[93]	"Asia"	"Asia"
##	[95]	"Asia"	"Asia"
##	[97]	"Asia"	"Asia"
##	[99]	"Asia"	"Asia"
##	[101]	"Asia"	"Asia"
##	[103]	"Asia"	"Asia"

```
## [105] "Asia" "Asia"
## [107] "Asia" "Asia"
## [109] "Asia" "Asia"
## [111] "Europe" "Europe"
## [113] "Europe" "Europe"
## [115] "Europe" "Europe"
## [117] "Europe" "Europe"
## [119] "Europe" "Europe"
## [121] "Europe" "Europe"
## [123] "Europe" "Europe"
## [125] "Asia" "Europe"
## [127] "Europe" "Europe"
## [129] "Europe" "Europe"
## [131] "Europe" "Europe"
## [133] "Europe" "Europe"
## [135] "Oceania" "Oceania"
## [137] "Oceania" "Oceania"
## [139] "Asia" "Asia"
## [141] "Asia" "Europe"
## [143] "Europe" "Europe"
## [145] "Europe" "Asia"
## [147] "Asia" "Asia"
## [149] "Asia" "Asia"
## [151] "Europe" "Europe"
## [153] "Europe" "Europe"
## [155] "Africa" "Asia"
## [157] "South America" "Asia"
## [159] "Asia" "Antarctica"
## [161] "Asia" "Asia"
## [163] "Africa" "Africa"
## [165] "Africa" "Africa"
## [167] "Africa" "Africa"
## [169] "Africa" "Africa"
## [171] "Europe" "Europe"
## [173] "Europe" "Europe"
## [175] "Europe" "North America"
## [177] "Africa"
```

```
world$continent[world$continent == "North America"]
```

```
## [1] "North America" "North America" "North America" "North America"
## [5] "North America" "North America" "North America" "North America"
## [9] "North America" "North America" "North America" "North America"
## [13] "North America" "North America" "North America" "North America"
## [17] "North America" "North America"
```

```
world %>%
  filter(continent == "North America")
```

```
## Simple feature collection with 18 features and 10 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -171.7911 ymin: 7.220541 xmax: -12.20855 ymax: 83.64513
## Geodetic CRS:   WGS 84
## # A tibble: 18 × 11
##   iso_a2 name_long continent region_un subregion type area_km2 pop lifeExp
## * <chr> <chr>      <chr>      <chr>      <chr>      <chr>      <dbl> <dbl> <dbl>
## 1 CA      Canada      North Am... Americas Northern... Sove...  1.00e7 3.55e7  82.0
## 2 US      United St... North Am... Americas Northern... Coun...  9.51e6 3.19e8  78.8
## 3 HT      Haiti       North Am... Americas Caribbean Sove...  2.85e4 1.06e7  62.8
## 4 DO      Dominican... North Am... Americas Caribbean Sove...  4.82e4 1.04e7  73.5
## 5 BS      Bahamas    North Am... Americas Caribbean Sove...  1.56e4 3.82e5  75.4
## 6 GL      Greenland  North Am... Americas Northern... Coun...  2.21e6 5.63e4   NA
## 7 MX      Mexico     North Am... Americas Central ... Sove...  1.97e6 1.24e8  76.8
## 8 PA      Panama     North Am... Americas Central ... Sove...  7.53e4 3.90e6  77.6
## 9 CR      Costa Rica North Am... Americas Central ... Sove...  5.38e4 4.76e6  79.4
## 10 NI     Nicaragua North Am... Americas Central ... Sove...  1.30e5 6.01e6  74.9
## 11 HN     Honduras  North Am... Americas Central ... Sove...  1.14e5 8.81e6  73.2
## 12 SV     El Salvad... North Am... Americas Central ... Sove...  2.09e4 6.28e6  73.0
## 13 GT     Guatemala North Am... Americas Central ... Sove...  1.09e5 1.59e7  72.9
## 14 BZ     Belize     North Am... Americas Central ... Sove...  2.20e4 3.52e5  70.0
## 15 PR     Puerto Ri... North Am... Americas Caribbean Depe...  9.22e3 3.53e6  79.4
## 16 JM     Jamaica    North Am... Americas Caribbean Sove...  1.25e4 2.86e6  75.7
## 17 CU     Cuba       North Am... Americas Caribbean Sove...  1.15e5 1.14e7  79.4
## 18 TT     Trinidad ... North Am... Americas Caribbean Sove...  7.74e3 1.35e6  70.4
## # i 2 more variables: gdpPercap <dbl>, geom <MULTIPOLYGON [°]>
```

```
world[world$continent == "North America" & world$type == "Sovereign country", ]
```

```
## Simple feature collection with 15 features and 10 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -140.9978 ymin: 7.220541 xmax: -52.6481 ymax: 83.23324
## Geodetic CRS:   WGS 84
## # A tibble: 15 × 11
##   iso_a2 name_long continent region_un subregion type area_km2    pop lifeExp
##   <chr>  <chr>      <chr>    <chr>    <chr>    <chr>    <dbl>  <dbl>  <dbl>
## 1 CA    Canada      North Am... Americas Northern... Sove...  1.00e7 3.55e7   82.0
## 2 HT    Haiti        North Am... Americas Caribbean Sove...  2.85e4 1.06e7   62.8
## 3 DO    Dominican... North Am... Americas Caribbean Sove...  4.82e4 1.04e7   73.5
## 4 BS    Bahamas      North Am... Americas Caribbean Sove...  1.56e4 3.82e5   75.4
## 5 MX    Mexico        North Am... Americas Central ... Sove...  1.97e6 1.24e8   76.8
## 6 PA    Panama        North Am... Americas Central ... Sove...  7.53e4 3.90e6   77.6
## 7 CR    Costa Rica    North Am... Americas Central ... Sove...  5.38e4 4.76e6   79.4
## 8 NI    Nicaragua     North Am... Americas Central ... Sove...  1.30e5 6.01e6   74.9
## 9 HN    Honduras      North Am... Americas Central ... Sove...  1.14e5 8.81e6   73.2
## 10 SV   El Salvad... North Am... Americas Central ... Sove...  2.09e4 6.28e6   73.0
## 11 GT    Guatemala     North Am... Americas Central ... Sove...  1.09e5 1.59e7   72.9
## 12 BZ    Belize        North Am... Americas Central ... Sove...  2.20e4 3.52e5   70.0
## 13 JM    Jamaica       North Am... Americas Caribbean Sove...  1.25e4 2.86e6   75.7
## 14 CU    Cuba          North Am... Americas Caribbean Sove...  1.15e5 1.14e7   79.4
## 15 TT    Trinidad ... North Am... Americas Caribbean Sove...  7.74e3 1.35e6   70.4
## # i 2 more variables: gdpPercap <dbl>, geom <MULTIPOLYGON [°]>
```

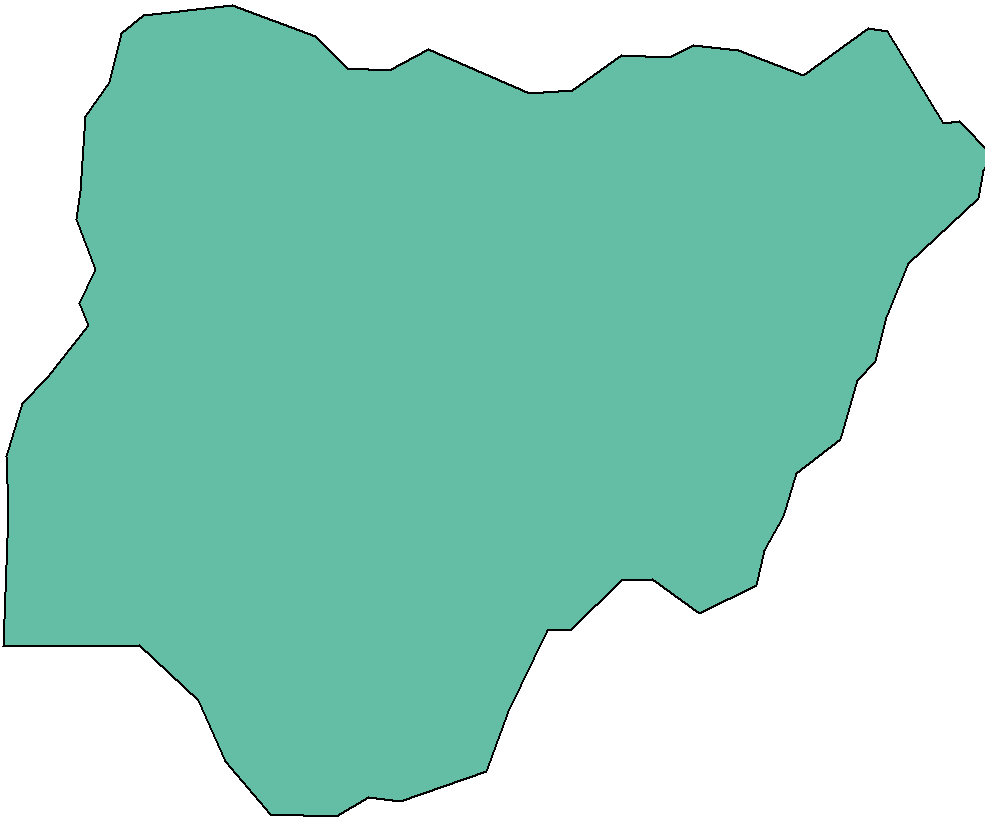
```
world %>%
  filter(continent == "North America", type == "Sovereign country")
```

```
## Simple feature collection with 15 features and 10 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -140.9978 ymin: 7.220541 xmax: -52.6481 ymax: 83.23324
## Geodetic CRS:   WGS 84
## # A tibble: 15 × 11
##   iso_a2 name_long continent region_un subregion type area_km2 pop lifeExp
## * <chr> <chr>      <chr>      <chr>      <chr>      <chr>      <dbl> <dbl> <dbl>
## 1 CA      Canada      North Am... Americas Northern... Sove...  1.00e7 3.55e7  82.0
## 2 HT      Haiti       North Am... Americas Caribbean Sove...  2.85e4 1.06e7  62.8
## 3 DO      Dominican... North Am... Americas Caribbean Sove...  4.82e4 1.04e7  73.5
## 4 BS      Bahamas    North Am... Americas Caribbean Sove...  1.56e4 3.82e5  75.4
## 5 MX      Mexico     North Am... Americas Central ... Sove...  1.97e6 1.24e8  76.8
## 6 PA      Panama     North Am... Americas Central ... Sove...  7.53e4 3.90e6  77.6
## 7 CR      Costa Rica North Am... Americas Central ... Sove...  5.38e4 4.76e6  79.4
## 8 NI      Nicaragua North Am... Americas Central ... Sove...  1.30e5 6.01e6  74.9
## 9 HN      Honduras   North Am... Americas Central ... Sove...  1.14e5 8.81e6  73.2
## 10 SV     El Salvad... North Am... Americas Central ... Sove...  2.09e4 6.28e6  73.0
## 11 GT     Guatemala North Am... Americas Central ... Sove...  1.09e5 1.59e7  72.9
## 12 BZ     Belize     North Am... Americas Central ... Sove...  2.20e4 3.52e5  70.0
## 13 JM     Jamaica    North Am... Americas Caribbean Sove...  1.25e4 2.86e6  75.7
## 14 CU     Cuba       North Am... Americas Caribbean Sove...  1.15e5 1.14e7  79.4
## 15 TT     Trinidad ... North Am... Americas Caribbean Sove...  7.74e3 1.35e6  70.4
## # i 2 more variables: gdpPercap <dbl>, geom <MULTIPOLYGON [°]>
```

To create a map using R basic plot

```
plot(nigeria["name_long"])
```

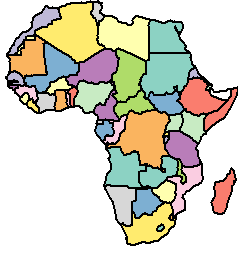
name_long



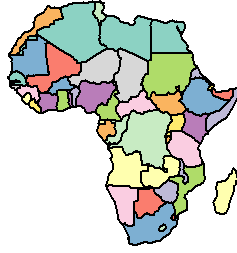
```
plot(Africa)
```

```
## Warning: plotting the first 9 out of 10 attributes; use max.plot = 10 to plot  
## all
```

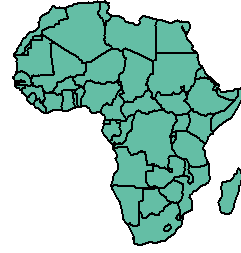
iso_a2



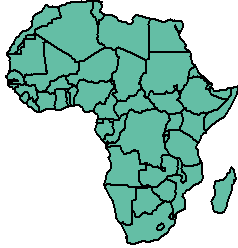
name_long



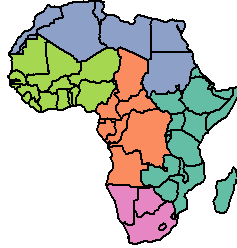
continent



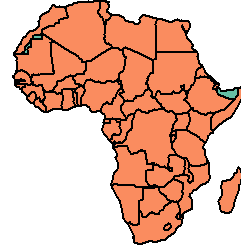
region_un



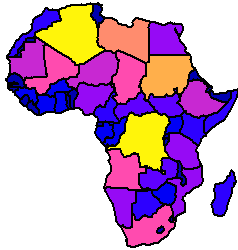
subregion



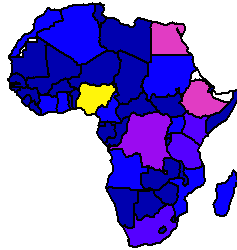
type



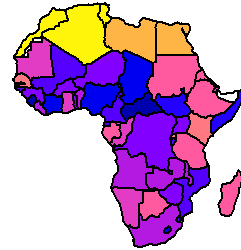
area_km2



pop



lifeExp



```
plot(Africa["geom"])
```



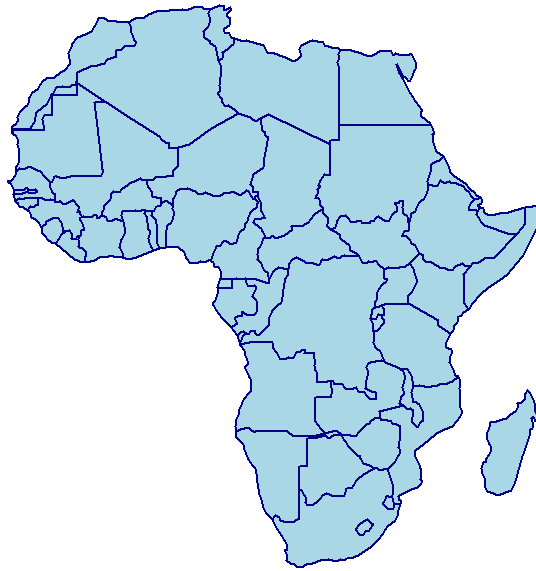

```
plot(st_geometry(Africa))
```



To modify the plot

```
plot(st_geometry(Africa),  
     main = "Map of Africa",  
     cex.main = 2,          # Title size  
     col = "lightblue",     # Fill color  
     border = "darkblue")  # Border color
```

Map of Africa



Adding labels on the map

```
plot(st_geometry(Africa), main = "African Countries")  
text(st_coordinates(st_centroid(Africa)), labels = Africa$name_long, cex = 0.6)
```

```
## Warning: st_centroid assumes attributes are constant over geometries
```

African Countries



Increasing the size of the map, title name and label name

```
plot(st_geometry(Africa),  
     main = "Map of Africa", cex.main = 5)  
text(st_coordinates(st_centroid(Africa)),  
     labels = Africa$name_long, cex = 2.5)
```

```
## Warning: st_centroid assumes attributes are constant over geometries
```

Map of Africa



```
plot(st_geometry(Africa),
     main = "Map of Africa",
     cex.main = 5,
     col = "lightblue",
     border = "darkgreen")
text(st_coordinates(st_centroid(Africa)),
     labels = Africa$name_long,
     cex = 2.5,
     col = "black")
```

```
## Warning: st_centroid assumes attributes are constant over geometries
```

Map of Africa



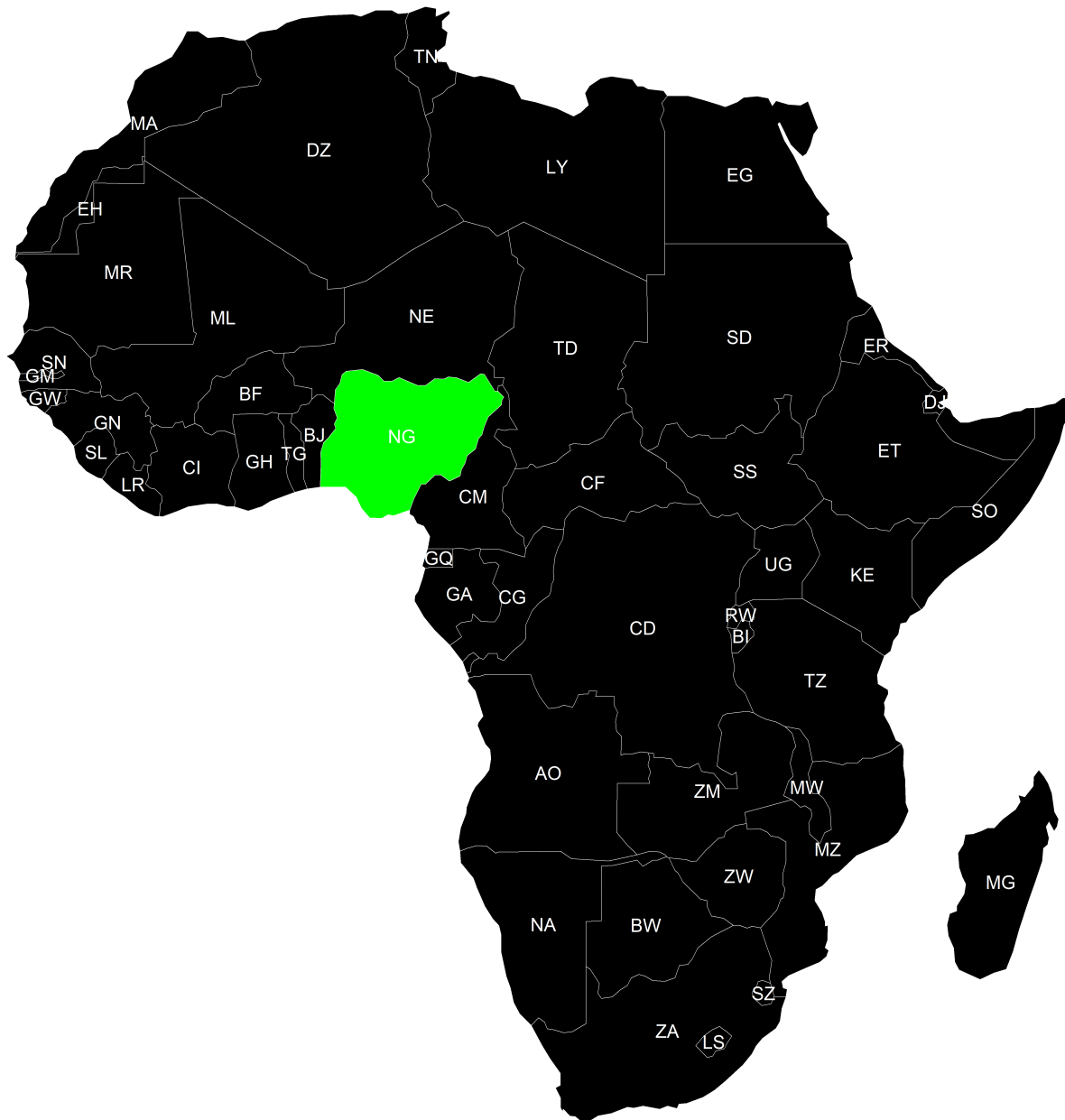
To highlight Nigeria using a different color

```
country_colors <- ifelse(Africa$name_long == "Nigeria", "green", "black")

plot(
  st_geometry(Africa),
  main = "Map of Africa",
  cex.main = 5,
  col = country_colors,
  border = "white")
text(st_coordinates(st_centroid(Africa)),
  labels = Africa$iso_a2,
  cex = 2.6,
  col = "white"
)
```

```
## Warning: st_centroid assumes attributes are constant over geometries
```

Map of Africa



Highlighting two countries

```
congo_colors <- ifelse(Africa$name_long %in% c("Republic of the Congo", "Democratic Republic of
the Congo"),
  c("orange", "red")[match(Africa$name_long, c("Republic of the Congo", "Democratic Republic of
the Congo"))],
  "lightblue"
)

plot(
  st_geometry(Africa),
  main = "Map of Africa",
  cex.main = 5,
  col = congo_colors,
  border = "black"
)

text(st_coordinates(st_centroid(Africa)),
  labels = Africa$name_long,
  cex = 2.0,
  col = "black"
)
```

```
## Warning: st_centroid assumes attributes are constant over geometries
```

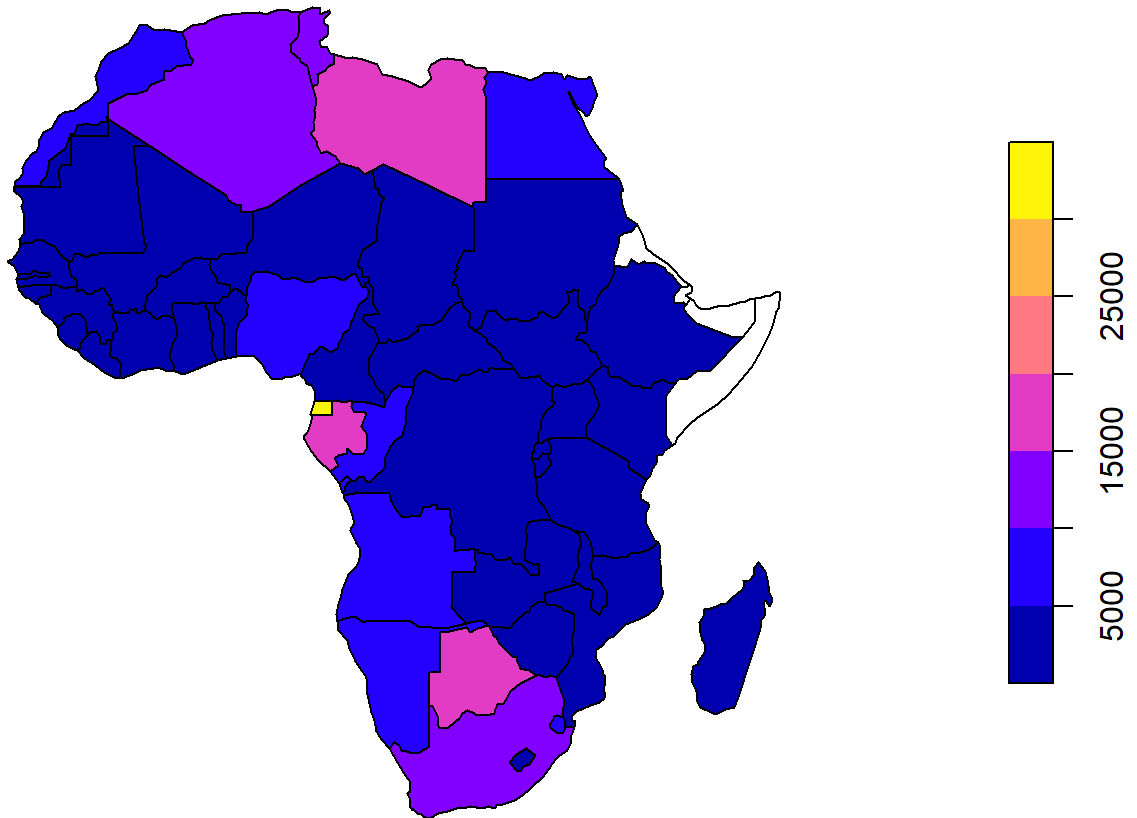
Map of Africa



Choropleth map - GDP by continent, using color gradients to show differences in economic output.

```
plot(Africa["gdpPercap"], main = "GDP per Capita in Africa")
```

GDP per Capita in Africa

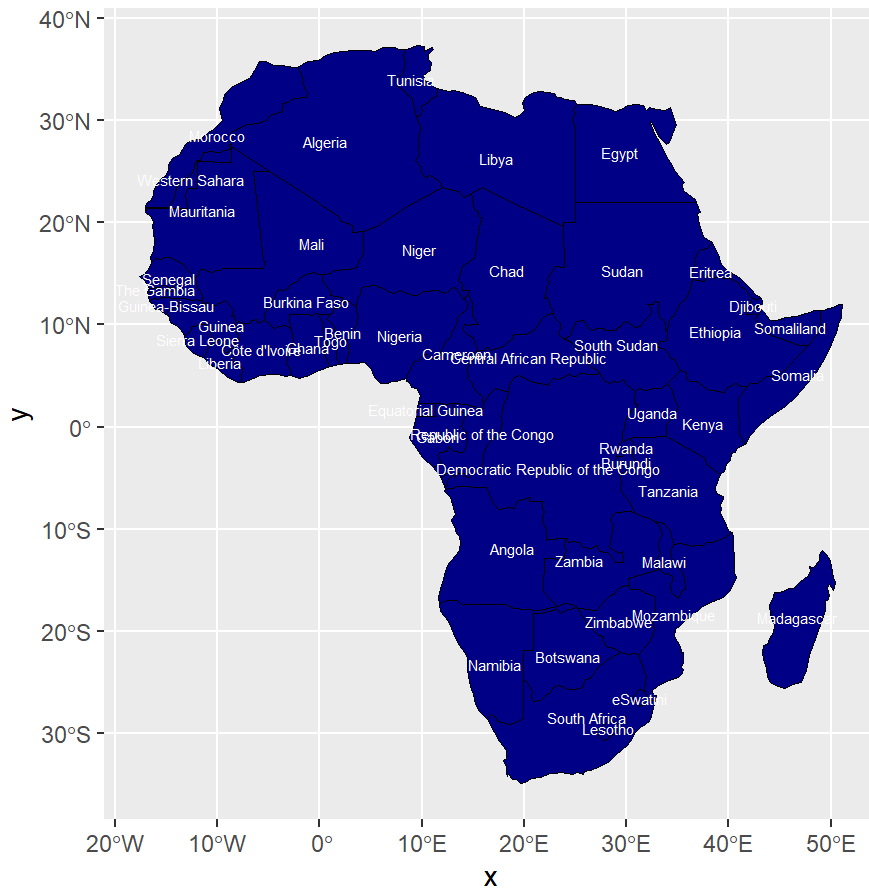


Creating map using ggplot2

```
ggplot(data = Africa) +  
  geom_sf(fill = "darkblue", color = "black") +  
  geom_sf_text(aes(label = name_long), size = 2, color = "white") +  
  ggtitle("Map of Africa")
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not  
## give correct results for longitude/latitude data
```

Map of Africa



```
ggplot(data = Africa) +  
  geom_sf(fill = "darkblue", color = "black") +  
  geom_sf_text(aes(label = name_long), size = 10, color = "white") +  
  ggtitle("Map of Africa")+  
  theme(plot.title = element_text(  
    size = 28, face = "bold", hjust = 0.5))
```

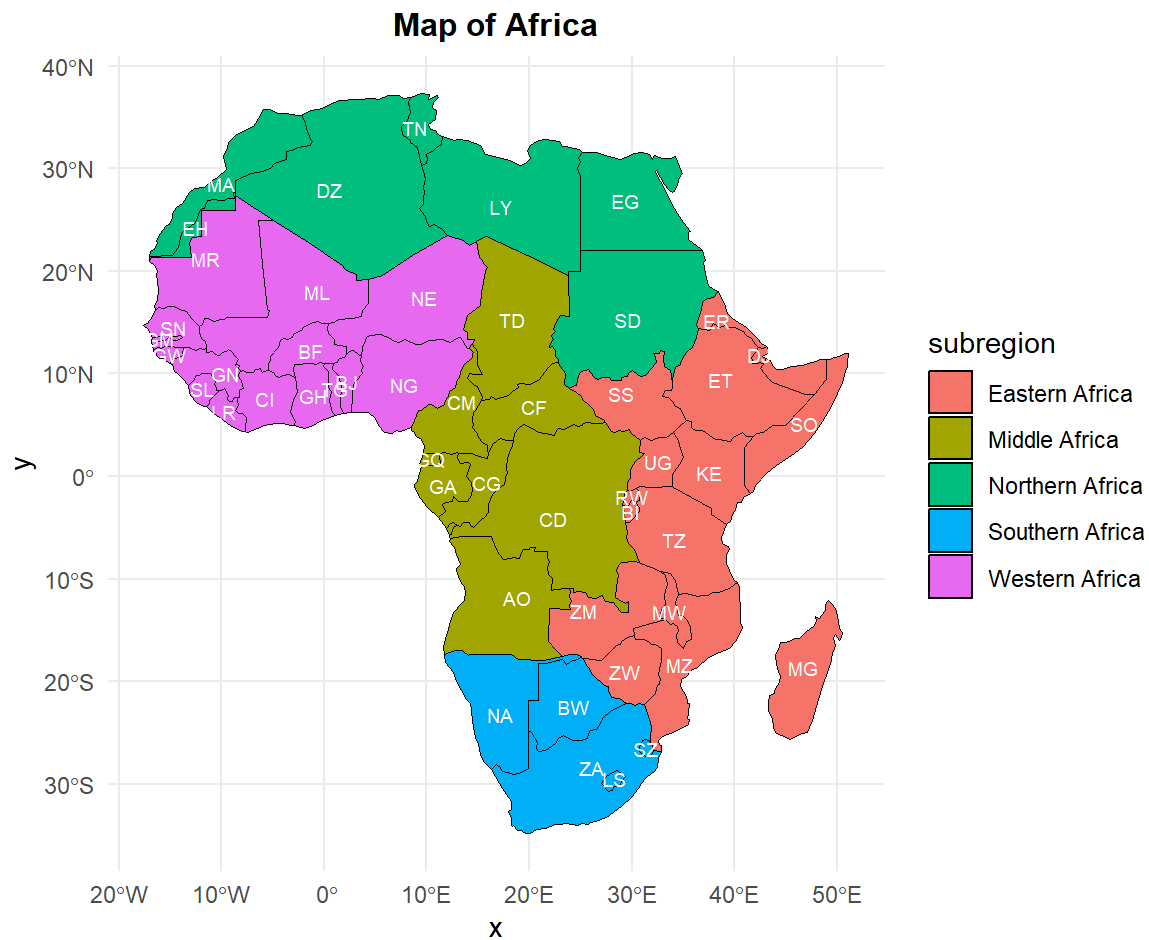
```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not  
## give correct results for longitude/latitude data
```



```
ggplot(data = Africa) +
  geom_sf(aes(fill = subregion), color = "black") +
  geom_sf_text(aes(label = iso_a2), size = 2.5, color = "white") +
  ggtitle("Map of Africa") +
  theme_minimal() +
  theme(plot.title = element_text(size = 12, face = "bold", hjust = 0.5))
```

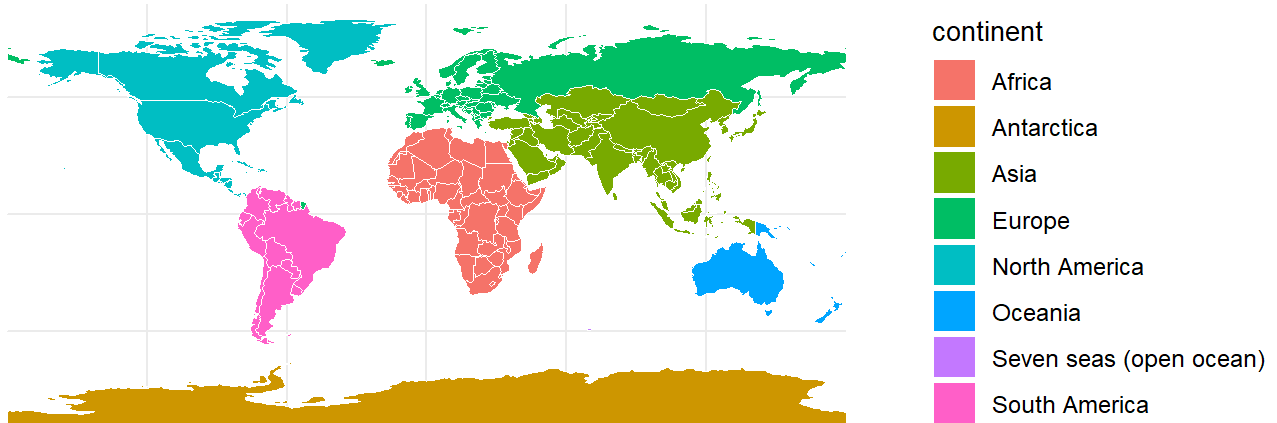
```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_text()`).
```



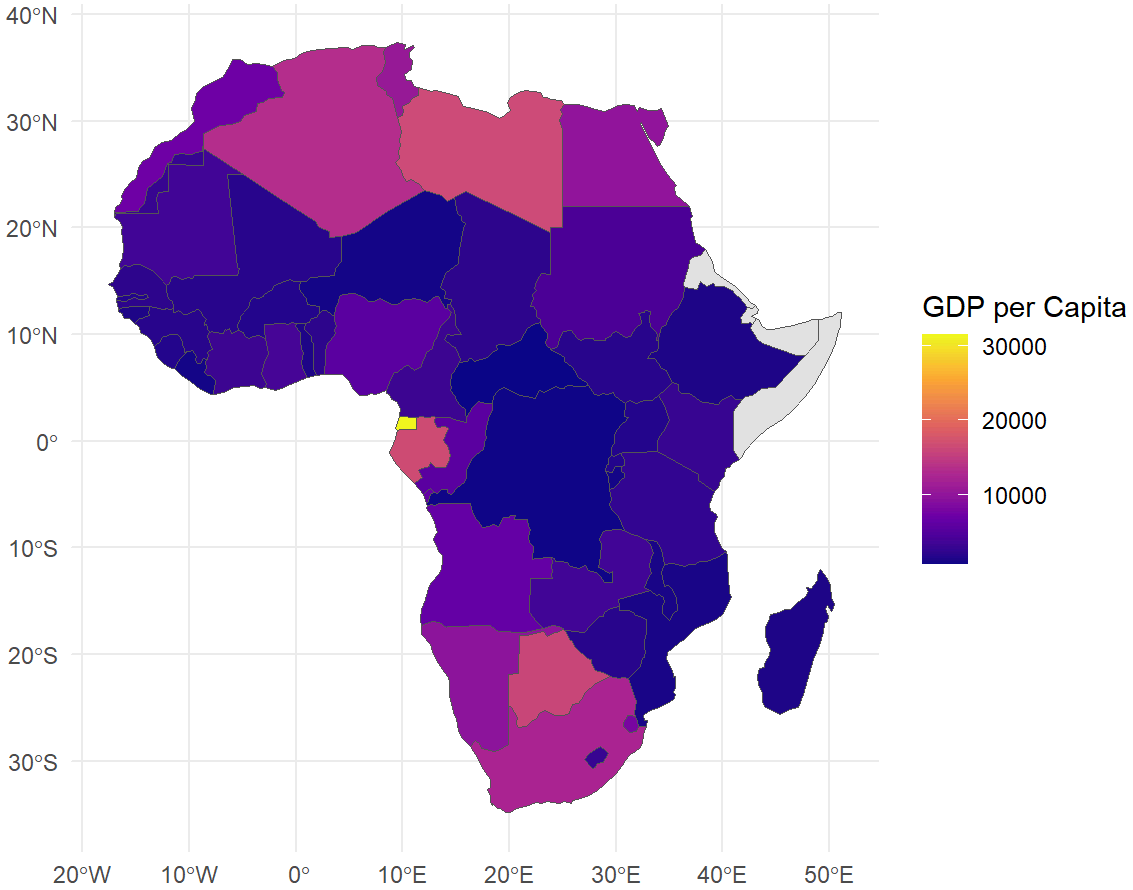
```
ggplot(data = world) +
  geom_sf(aes(fill = continent), color = "white") +
  ggtitle("World Map by Continent") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    legend.title = element_text(size = 10),
    legend.text = element_text(size = 9)
  )
```

World Map by Continent



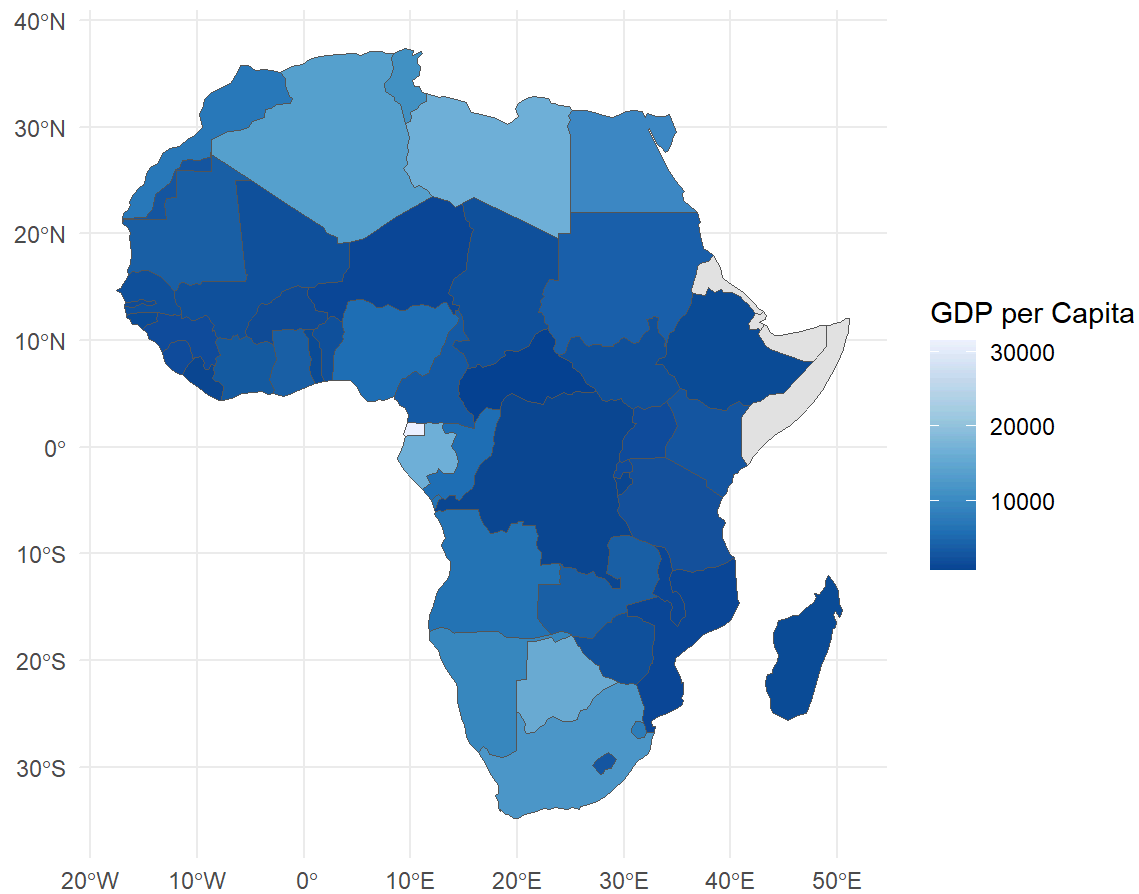
```
ggplot(Africa) +  
  geom_sf(aes(fill = gdpPercap)) +  
  scale_fill_viridis_c(option = "C", na.value = "grey90", name = "GDP per Capita") +  
  theme_minimal() +  
  labs(title = "GDP per Capita in Africa")
```

GDP per Capita in Africa



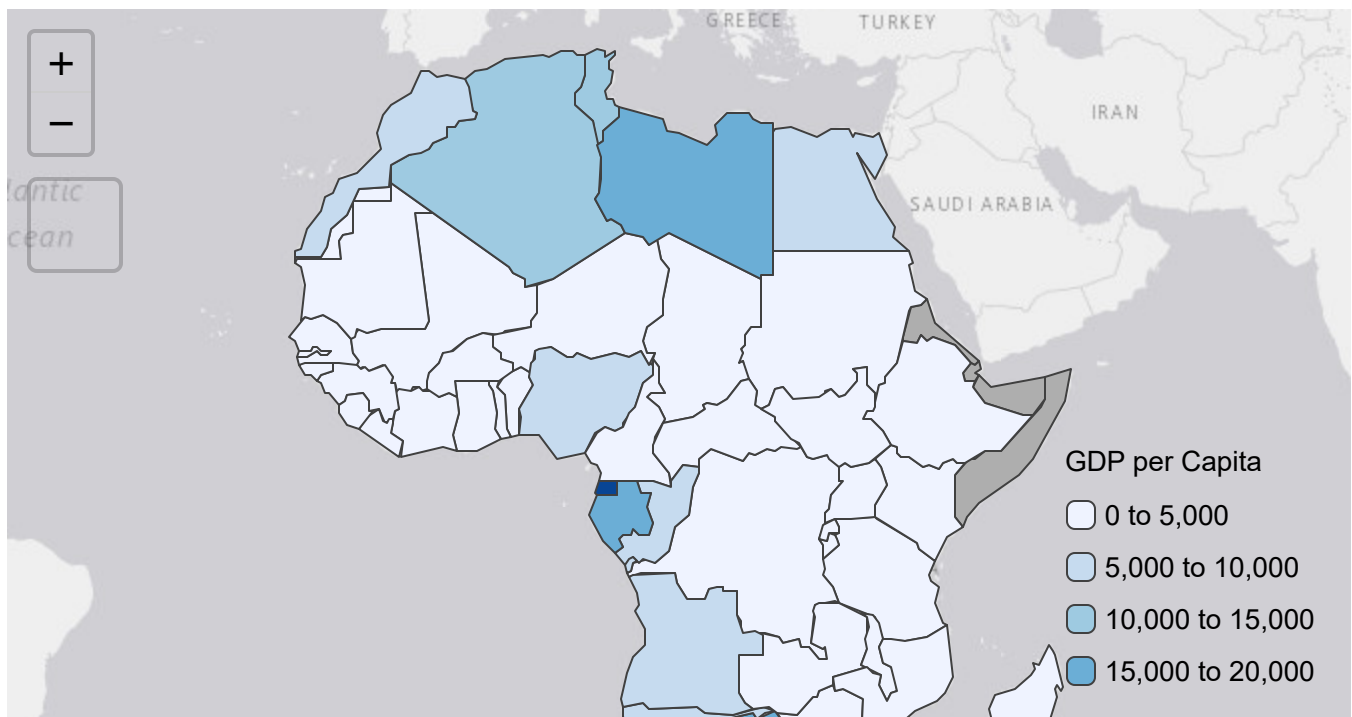
```
ggplot(Africa) +
  geom_sf(aes(fill = gdpPercap)) +
  scale_fill_distiller(palette = "Blues", na.value = "grey90", name = "GDP per Capita") +
  theme_minimal() +
  labs(title = "GDP per Capita in Africa")
```

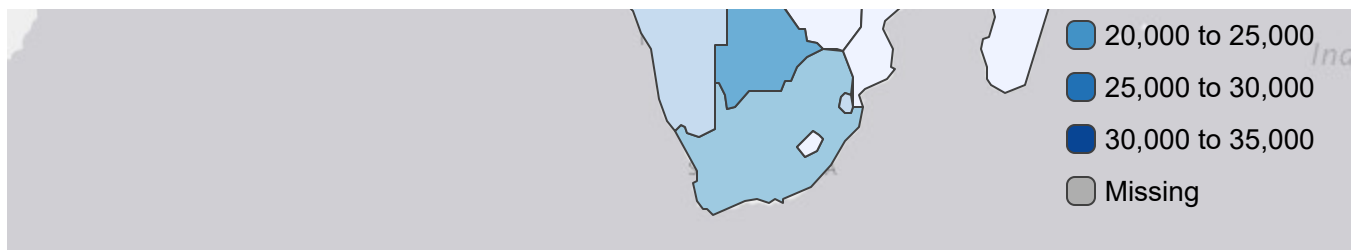

GDP per Capita in Africa



tmap

```
tmap_mode("view") # Interactive map
tm_shape(Africa) +
  tm_polygons("gdpPercap", palette = "Blues", title = "GDP per Capita")
```



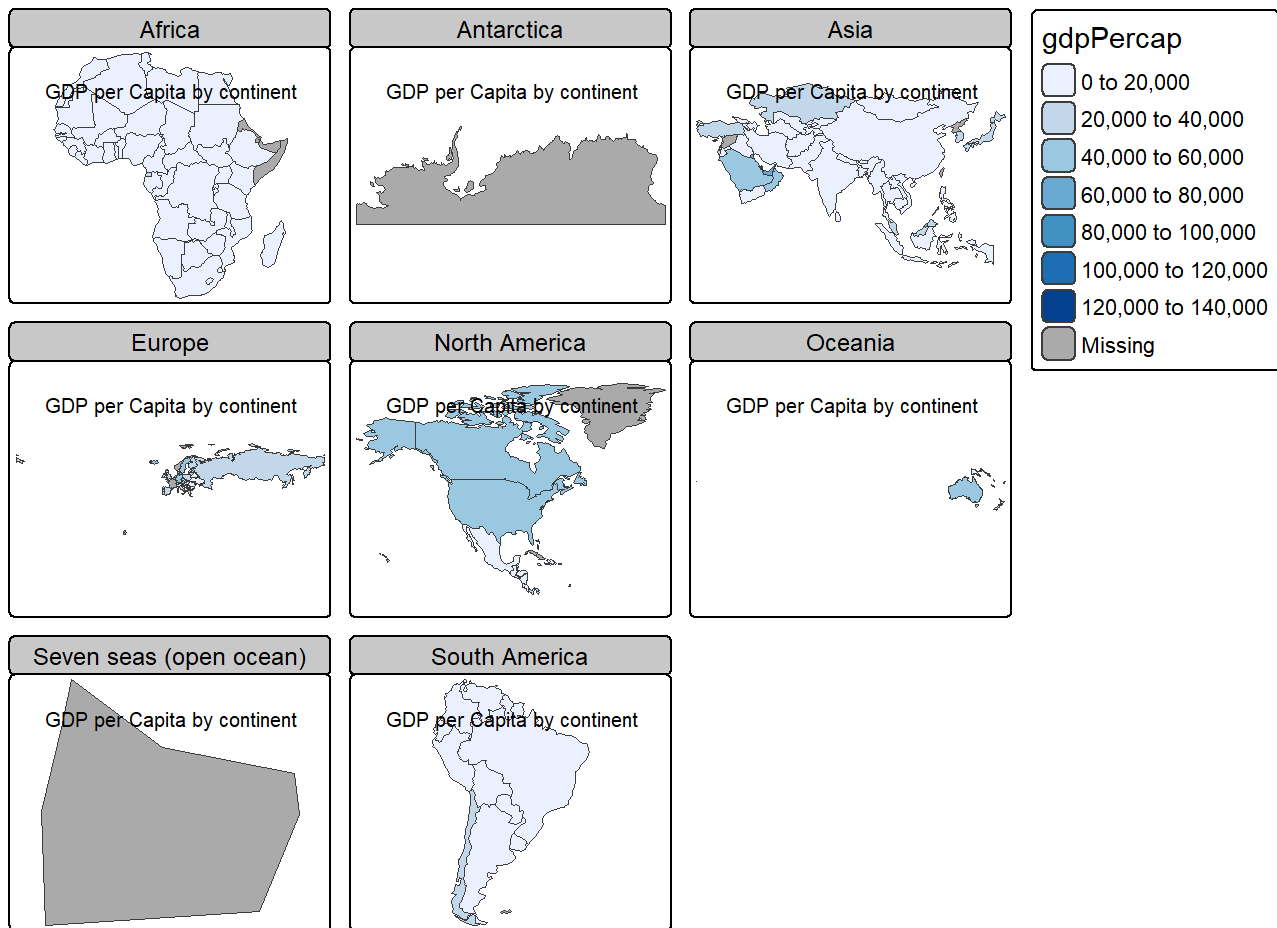


```
tmap_mode("plot")
```

```
## i tmap mode set to "plot".
```

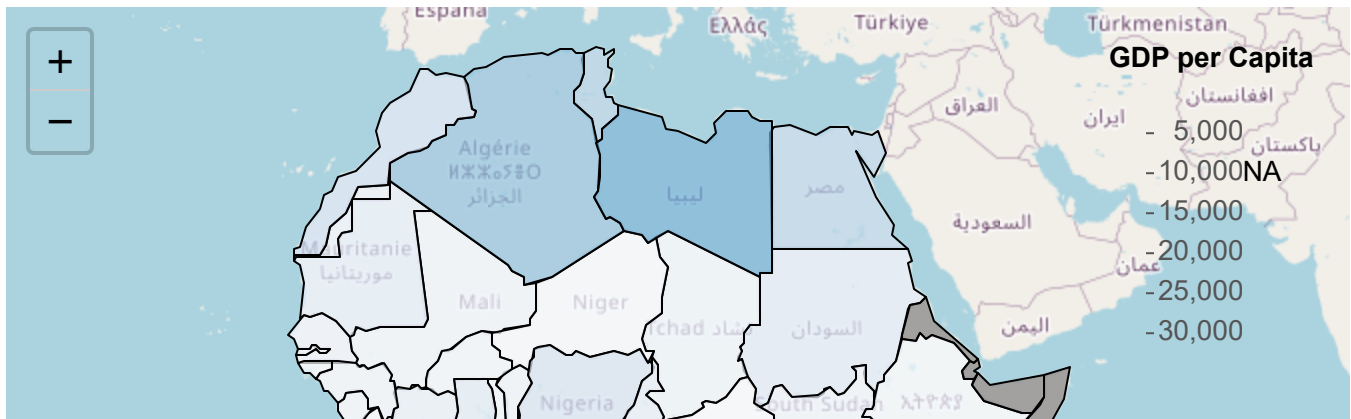
```
tm_shape(world) +
  tm_polygons("gdpPercap", palette = "Blues") +
  tm_facets(by = "continent", ncol = 3) +
  tm_layout(title = "GDP per Capita by continent")
```

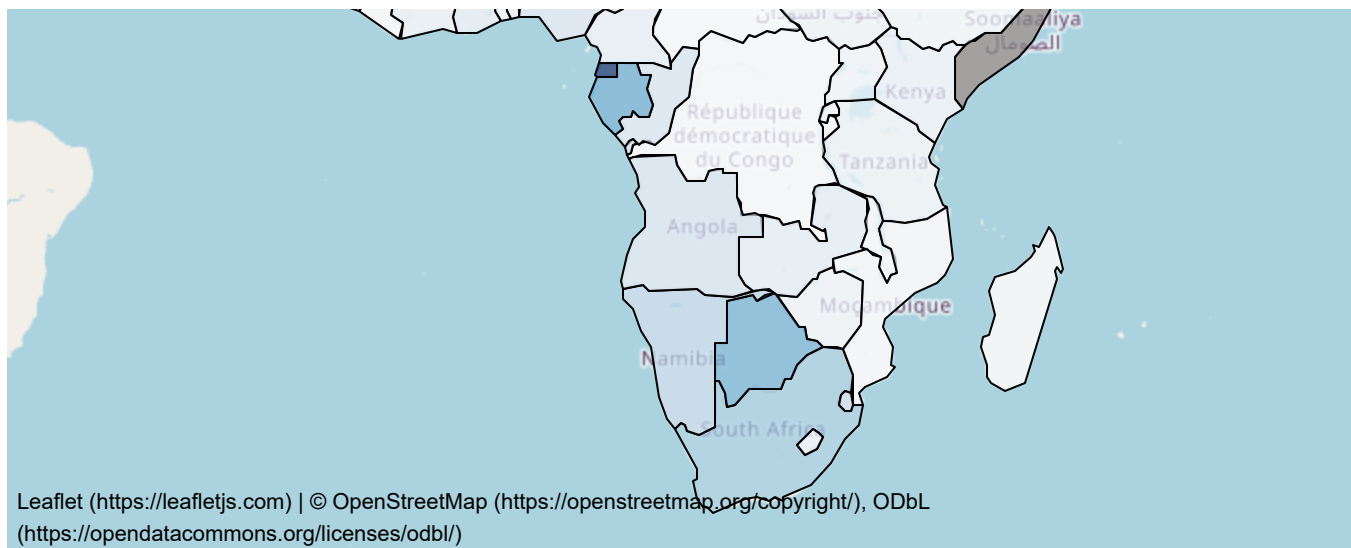
```
##
## — tmap v3 code detected —————
## [v3->v4] `tm_tm_polygons()`: migrate the argument(s) related to the scale of
## the visual variable `fill` namely 'palette' (rename to 'values') to fill.scale
## = tm_scale(<HERE>).[v3->v4] `tm_layout()`: use `tm_title()` instead of `tm_layout(title = )`
[tip] Consider a suitable map projection, e.g. by adding `+ tm_crs("auto")`. [cols4all] color pal
ettes: use palettes from the R package cols4all. Run
## `cols4all::c4a_gui()` to explore them. The old palette name "Blues" is named
## "brewer.blues" Multiple palettes called "blues" found: "brewer.blues", "matplotlib.blues". The
first one, "brewer.blues", is returned.
## [plot mode] fit legend/component: Some legend items or map components do not
## fit well, and are therefore rescaled.
## i Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```



Leaflet

```
leaflet(Africa) %>%
  addTiles() %>%
  addPolygons(fillColor = ~colorNumeric("Blues", gdpPercap)(gdpPercap),
    color = "black",
    weight = 1,
    opacity = 1,
    fillOpacity = 0.7,
    label = ~paste(name_long, "<br>", "GDP per Capita:", round(gdpPercap, 0))) %>%
  addLegend(pal = colorNumeric("Blues", Africa$gdpPercap),
    values = Africa$gdpPercap,
    title = "GDP per Capita")
```





```
# Create a ggplot map (2D version)
africa_map <- ggplot(Africa) +
  geom_sf(aes(fill = gdpPercap), color = "black", size = 0.2) +
  scale_fill_viridis(option = "C", na.value = "grey90", name = "GDP per Capita") +
  theme_minimal() +
  labs(title = "GDP per Capita in Africa")
```

```
# Convert to 3D using rayshader
plot_gg(africa_map,
  multicore = TRUE,
  width = 5,
  height = 5,
  scale = 300,          # Adjust this for exaggeration
  window_size = c(1000, 800),
  zoom = 0.6,
  phi = 40,
  theta = 135,
  fov = 60)
```

```
# Save the 3D snapshot
# render_snapshot("africa_gdp_3d.png")
```